

Attention Is All You Need ? Overview

We propose the Transformer, a model architecture relying entirely on an attention mechanism to draw global dependencies between input and output, dispensing with recurrence and convolutions entirely.

Scaled Dot-Product Attention

The input consists of queries and keys of dimension d_k and values of dimension d_v . We compute the dot products of the query with all keys, divide each by $\sqrt{d_k}$, and apply a softmax function to obtain the weights on the values.

Multi-Head Attention and Results

Multi-head attention allows the model to jointly attend to information from different representation subspaces at different positions. On WMT 2014 English-to-German translation the Transformer achieves state-of-the-art BLEU scores.